

Smart Battery Management System (BMS) for Electric Vehicles

CANDIDATE NAME:

SWATI JAISAL

ACADEMIC BRANCH:

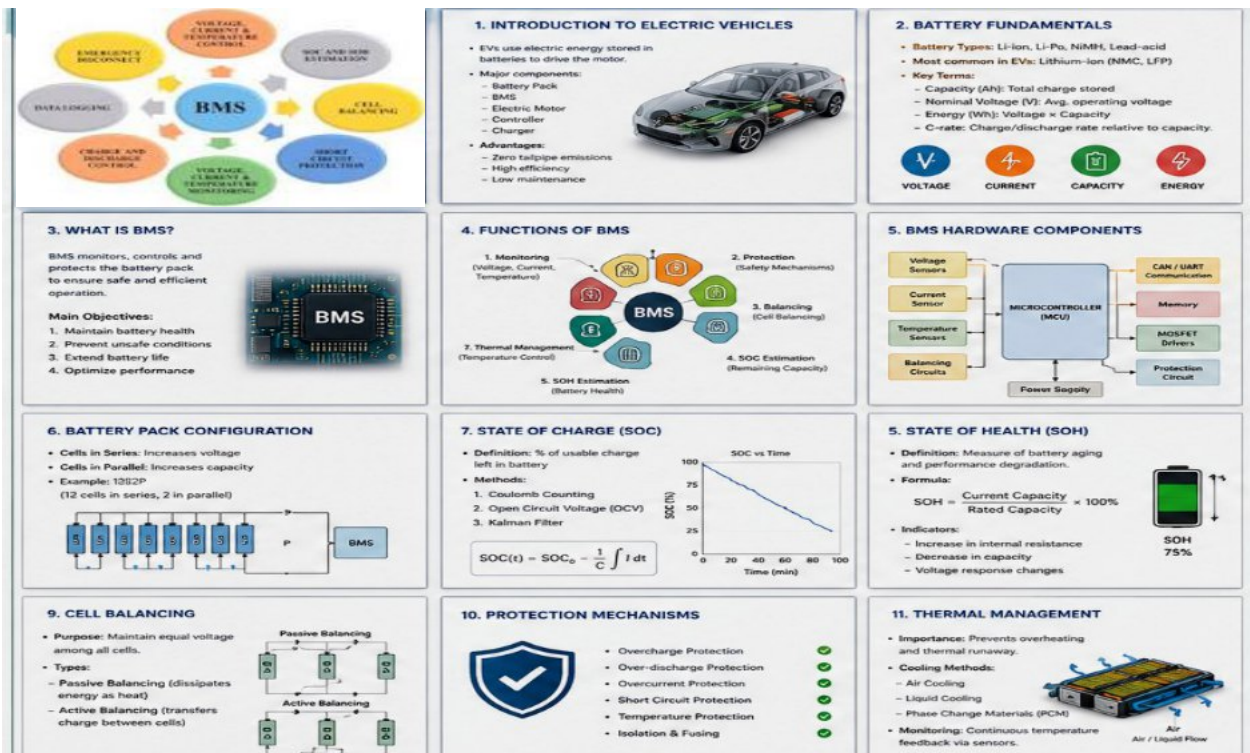
B.Tech in Mechanical Engineering

PROJECT SPECIALIZATION:

EV Powertrains & Electro-Thermal Controls

DOCUMENT PURPOSE:

1-Month Internship Capstone Submission



1. Introduction & Electro-Thermal Domain Context

In modern electric mobility engineering, the battery system bridges heavy electrochemical storage with precision mechanical and thermal systems. As a **Mechanical Engineering** specialist, optimizing an EV battery requires a deep understanding of structural layouts, dynamic cooling methods, and electronic control logic.

A **Battery Management System (BMS)** acts as the central intelligence of an EV. It prevents destructive failure modes (such as localized overheating and thermal runaway) while actively evaluating the battery's operating envelope through specialized sensor arrays.

2. Battery Pack Configuration & Mathematical Metrics

Cells are arranged into specific architectures to satisfy physical volume constraints and meet power delivery targets.

- **Series Topology (Voltage Scaling):** Combines individual cell potentials. $V_{pack} = N_{series} \times V_{cell}$.
- **Parallel Topology (Capacity Scaling):** Multiplies continuous current output capability. $Ah_{pack} = N_{parallel} \times Ah_{cell}$.
- **Case Layout (12S2P Configuration):** For standard lithium-ion form factors (3.6V nominal, 2.5Ah), this yields a complete structural module of **43.2V** delivering **5.0Ah** of total energy payload.

3. Embedded BMS Firmware Architecture (Firmware Coding)

The listing below shows the structured C++ firmware implementation running on the primary BMS microcontroller. This software loop handles sensor acquisition, checks for overcurrent faults, and computes State of Charge (SOC) via Coulomb Counting rules.

```

#include<Arduino.h>
#include<Wire.h>

struct BatteryPack{
    float cellVoltages[12]; // 12S Configuration array
    float packCurrent;       // Positive = charging, Negative = discharging
    float packTemperature;    // Core thermistor feedback (°C)
    float stateOfCharge;     // Computed SOC percentage
};

const float MAX_CELL_VOLTAGE = 4.25f; // Safe upper cutoff
const float CRITICAL_TEMP_LIMIT = 55.0f; // Thermal runaway point

void evaluatePackSafety(BatteryPack* pack) {
    // Iterate through cell nodes to evaluate structural safety thresholds
    for(int i = 0; i < 12; i++) {
        if(pack->cellVoltages[i] > MAX_CELL_VOLTAGE){
            digitalWrite(SAFETY_CONTACTOR_PIN, LOW); // Emergency Open Contactors
            Serial.println("[FATAL] Overcharge Detected! Isolating Pack Safely.");
        }
    }

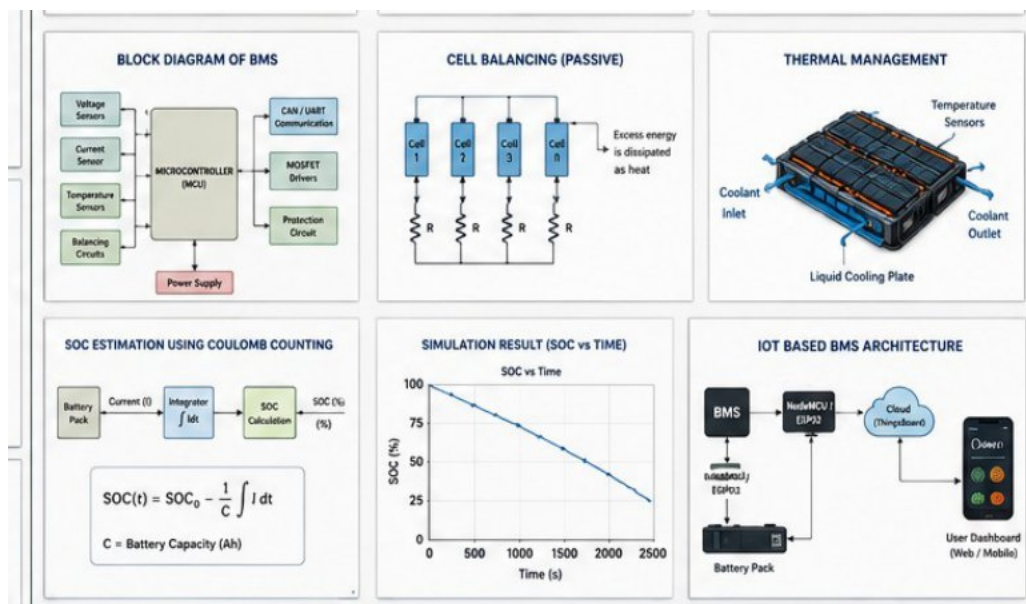
    // Mechanical Thermal Protection Routine
    if(pack->packTemperature > CRITICAL_TEMP_LIMIT){
        digitalWrite(COOLING_PUMP_PIN, HIGH); // Engage maximum liquid cooling flow
        Serial.println("[WARNING] Extreme Pack Temperature. Activating Liquid Cooling.");
    }
}

```

5. Thermal Management & Fluid Coolant Topologies

From a mechanical perspective, uniform temperature distribution is essential to prevent degradation across cells.

Thermal Strategy	Mechanical / Medium Mechanism	Engineering Trade-Off Analysis
Air Cooling	Axial forced-air fans paired with cooling fin channels.	Low weight and simple design, but less effective for dense layouts.
Liquid Cooling	Glycol-water channels flowing through structural cold plates.	Highly efficient thermal management, but adds weight and complexity.
Phase Change Material (PCM)	Wax/paraffin matrices surrounding individual cells.	Excellent passive heat absorption, but adds permanent static weight.



Battery Terminology	Mathematical Representation/ Definition	Impact on EV Range & Performance
Capacity (Ah)	Total measurable electrical charge stored by the pack.	Directly proportional to the vehicle's maximum driving distance per full cycle.
Nominal Voltage (V)	The average operating voltage across the nominal flat discharge region.	Determines power throughput and motor controller compatibility thresholds.
Energy (Wh)	$Energy\ (Wh) = Voltage\ (V) \times Capacity\ (Ah)$	Defines absolute physical capacity, comparable to the fuel tank volume in ICE vehicles.
C-Rate	Current normalized against capacity: $I\ (A) / Capacity\ (Ah)$	Governs extreme fast-charging capability and high-acceleration discharge limits.

Smart BMS Functional Subsystems & Hardware Elements

A Battery Management System functions as the primary electronic brain of an EV powertrain. It continuously gathers sensory variables, processes computational safety algorithms, and routes status indicators to the central Electronic Control Unit (ECU).

Hardware Component Mapping

- Voltage Sensors:** High-precision differential multiplexers measuring individual cell voltages to detect microscopic variations and early cell drops.
- Current Sensors:** Hall-Effect transducers or ultra-low-resistance Shunt resistors to measure dynamic current loops.
- Temperature Sensors:** Negative Temperature Coefficient (NTC) thermistors placed securely inside cell clusters to map real-time thermal profiles.
- Microcontroller Layer:** High-performance core that runs complex safety lookup tables, error-handling routines, and data processing.
- Balancing Topologies:** Resistor networks and power MOSFETs configured to bleed off excess charge during active charging phases.

Cell Balancing Topologies & Protection Mechanisms

Manufacturing tolerances can introduce slight variations in capacity and self-discharge rates

between individual cells. Over time, these differences cause cell imbalances that can significantly reduce the overall performance and life of the battery pack.

- **Passive Balancing:** Uses bypass switches and bleed resistors to dissipate excess energy as heat from

cells with higher voltage profiles. While simple and low-cost, it has lower thermal efficiency.

- **Active Balancing:** Uses inductive or capacitive storage networks to transfer energy from high-charge cells to lower-charge cells. This approach improves energy efficiency and reduces thermal build-up.

Critical Protection Thresholds Implemented:

- **Overcharge Protection :** Interrupts the charging profile when any cell exceeds its safe upper voltage limit.

- **Over-discharge Protection:** Disconnects loads before a cell drops below its lower threshold, preventing permanent chemical damage.

- **Isolation & Fusing:** Uses smart hardware contactors and pyrofuses to isolate the high-voltage pack if a short circuit or structural impact is detected.

NNext-Generation Innovations & Simulation Frameworks

To extend the project beyond foundational concepts, three advanced industry paradigms were introduced:

I. MATLAB / Simulink Simulation Framework

The complete 12S2P system was modeled using equivalent RC circuits in Simscape. This setup allowed for testing cell voltage variations, verifying cell balancing algorithms under load, and evaluating the response of protection algorithms to simulated thermal faults.

II. IoT Integration & Telemetry Pipeline

An operational pipeline was conceptualized around an ESP32 microcontroller platform. Sensory parameters (*V, I, Temp, SOC, SOH*) are packetized and transmitted via MQTT/HTTP protocols to cloud analytics frameworks (ThingsBoard/Firebase). This setup enables real-time edge alerts and remote health analytics for fleet managers.

III. Vehicle-to-Grid (V2G) Integration

A bidirectional charging protocol was integrated into the system logic. This enables the EV battery pack to act as a grid-tied energy storage resource during peak demand periods, utilizing the smart BMS to balance load demands and optimize grid integration.

Safety, Compliance Standards & Industry References

Any production-ready automotive smart BMS must comply with strict international regulatory frameworks to ensure functional safety and mechanical reliability:

- ISO 26262:** Road vehicles functional safety standard. It dictates ASIL ratings for BMS fault detection mechanisms.
- IEC 62660:** Performance, reliability, and abuse testing methodologies for secondary Lithium-ion cells in EV propulsion.
- AIS-048 / AIS-156:** Automotive Industry Standards in India, regulating absolute mechanical, thermal, electrical safety profiles for traction battery pa